

1. The following transformations are applied to the function  $f(x) = x^2 - 3$ :

2

- A vertical dilation with scale factor 3.
- Then a horizontal dilation with scale factor 2.
- Then a vertical translation down 1 unit.

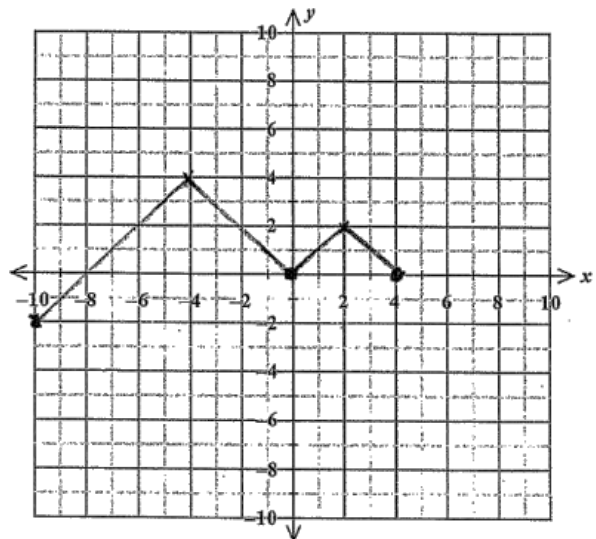
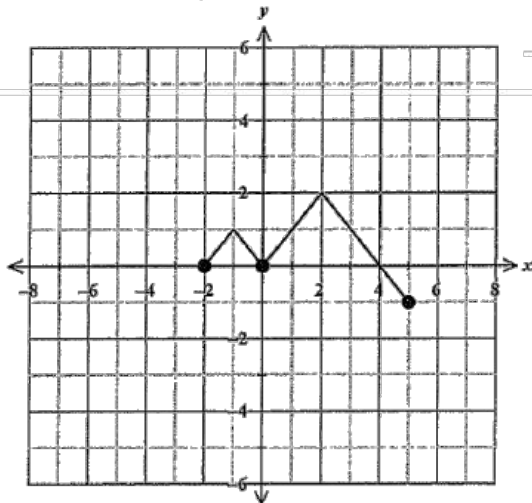
Which of the following is the equation of the transformed function  $g(x)$ ?

$$x^2 - 3 \rightarrow 3x^2 - 9 \rightarrow 3\left(\frac{1}{2}x\right)^2 - 9 \rightarrow \frac{3}{4}x^2 - 10$$

$$\frac{3}{4}x^2 - 9$$

2. Given the following function  $f(x)$ .  
Sketch  $2f\left(-\frac{1}{2}x\right)$  on the axis below.

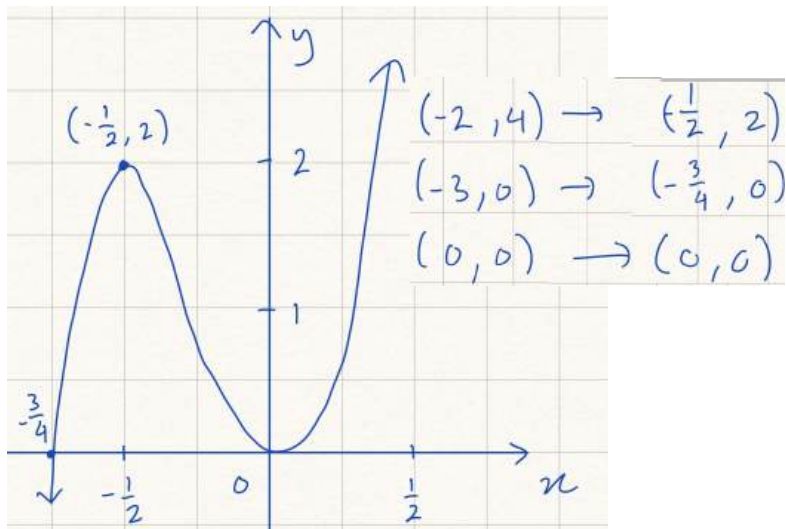
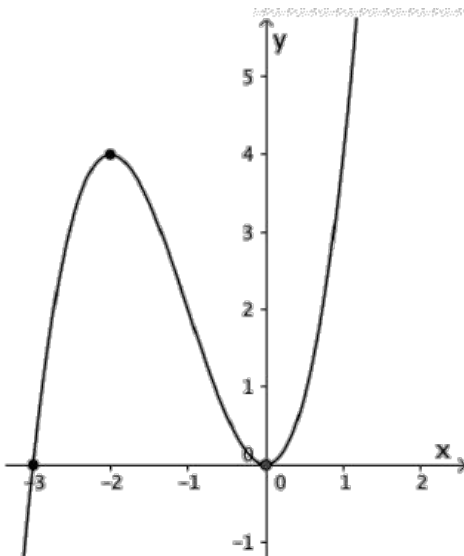
2



3. Consider the graph of  $y = f(x)$  as shown.

2

Sketch the graph of  $y = \frac{1}{2}f(4x)$  showing the  $x$ -intercepts and the coordinates of the turning points.



4. The point A (1, 4) lies on the graph  $y = f(x)$ . When the function is transformed into the graph  $y = -2f(x+1)$ , what are the co-ordinates of the point corresponding to point A?

2

↑ vertical dilation & reflection  
A (1, 4) → A (1, -8)

$f(x+1)$  is a horizontal translation  
A (1, -8) → A (0, -8) to the left

5. The graphs of  $y = f(x)$  and  $y = g(x)$  are shown. The function  $y = g(x)$  can be obtained by applying a series of transformations to  $y = f(x)$ . Which of the following is correct?

- A.  $g(x) = -f(x+4)$   
**B.  $g(x) = -f(2x+4)$**   
 C.  $g(x) = f(-2x) - 4$   
 D.  $g(x) = f(-x+2) - 4$

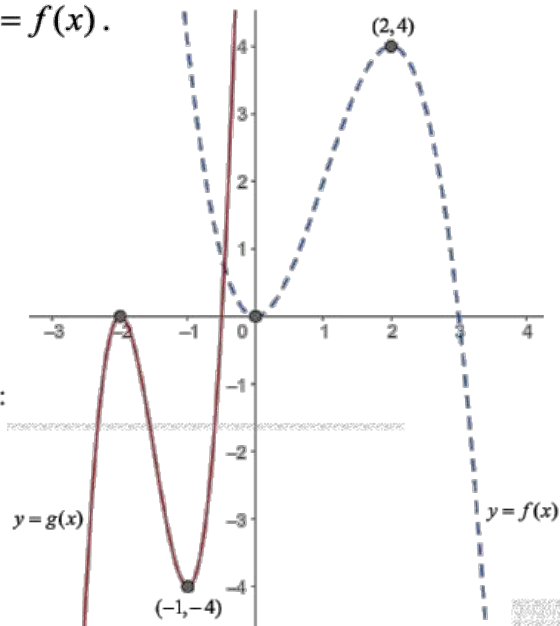
The graph of  $y = g(x)$  can be obtained from  $y = f(x)$  by applying:

A horizontal dilation by a factor of  $\frac{1}{2} \rightarrow f(2x)$

Translation by 2 units to left  $\rightarrow f(2(x+2))$

Reflection on x-axis:  $\rightarrow -f(2(x+2))$

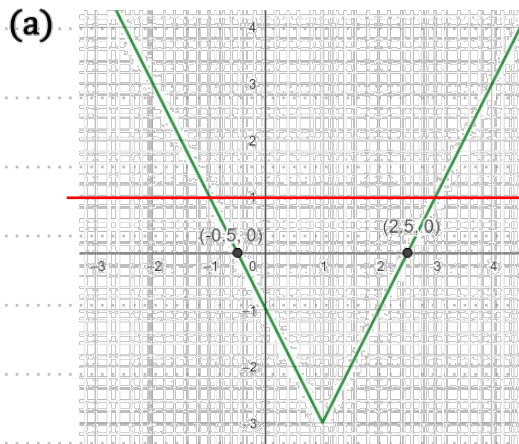
$g(x) = -f(2x+4) \Rightarrow B$



6. (a) Sketch the graph of  $y = 2|x - 1| - 3$ . Show all intercepts with the axes.  
 (b) Hence or otherwise solve the inequation  $2|x - 1| - 3 < -1$ .

2

2



- (b) Draw the line  $y = -1$  on the graph above. Solution is  $0 < x < 2$

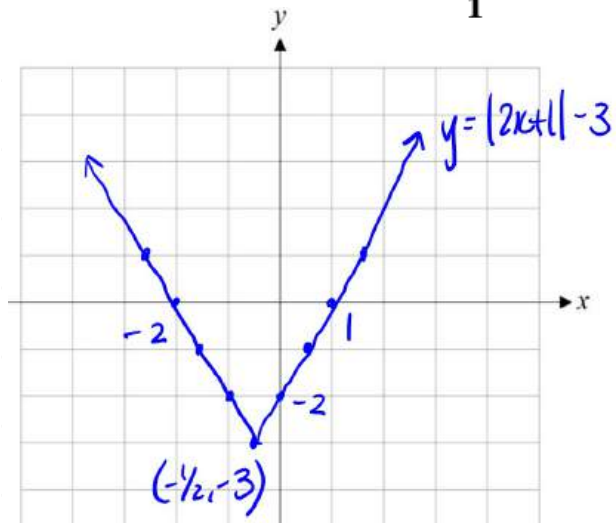
7. (a) On the axes below, graph the function  $y = |2x + 1| - 3$  labelling all important features. 2  
(b) Hence, or otherwise, solve  $|2x + 1| - 3 \leq 0$ . 1

(a)  $y = |2(x + \frac{1}{2})| - 3$

$\therefore$  Left  $\frac{1}{2}$ , down 3

$m = 2$

(b)  $-2 \leq x \leq 1$



8. The circle  $x^2 + y^2 - 6x + 8y - 11 = 0$  is transformed by a horizontal translation to the left by 4 units and a vertical translation up 3 units. 3  
What is the centre and radius of the new circle?

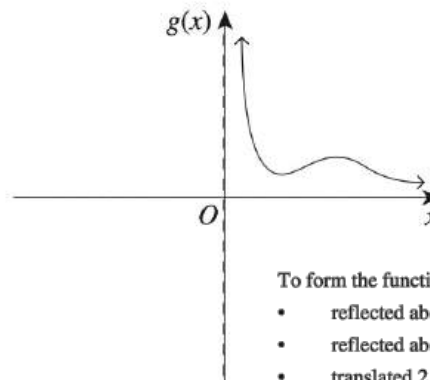
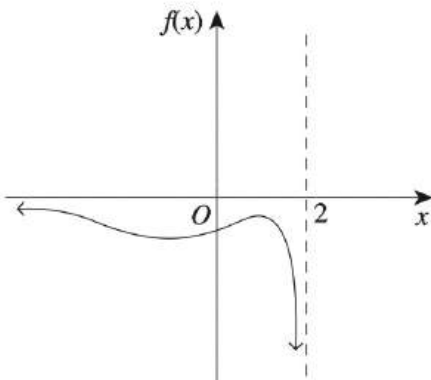
$$x^2 - 6x + (3)^2 + y^2 + 8y + (4)^2 = 11 + 9 + 16$$

$$(x - 3)^2 + (y + 4)^2 = 36$$

centre  $(3, -4)$   $r = 6$

$\therefore$  new centre  $(-1, -1)$   $r = 6$

9. The function  $y = f(x)$  is transformed to  $y = g(x)$ , as shown in the diagram.



To form the function  $g(x)$ ,  $f(x)$  has been:

- reflected about the y-axis, forming  $f(-x)$
- reflected about the x-axis, forming  $-f(-x)$
- translated 2 units to the left, forming  $-f(-(x - 2))$ .

Hence:

$$g(x) = -f(-(x - 2)) = -f(2 - x)$$

Which of the following equations best represents the transformed function?

- A.  $g(x) = -f(2 - x)$   
B.  $g(x) = -f(x - 2)$   
C.  $g(x) = f(2 - x)$   
D.  $g(x) = f(x - 2)$

10. The point  $A(4,2)$  lies on the graph of  $y = f(x)$ .

1

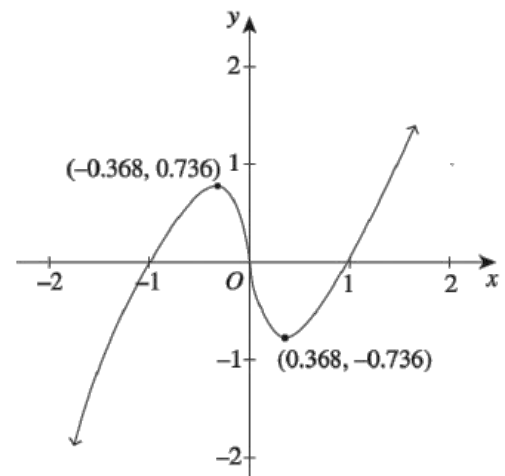
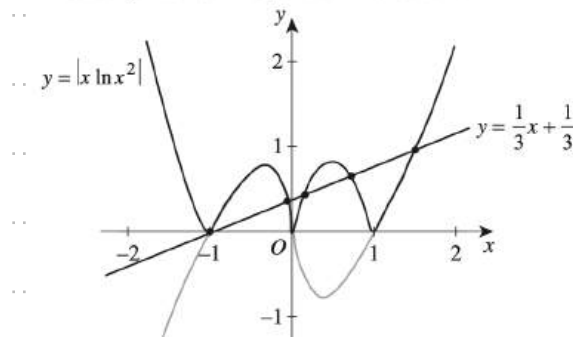
What are the coordinates of the point  $A'$ , the image point of  $A$ , on the graph of  $y = g(x)$  if  $g(x) = -f\left(\frac{x}{2}\right)$ ?

The graph of  $y = g(x)$  is obtained using the graph of  $y = f(x)$  through a reflection about the  $x$ -axis, and a horizontal dilation of factor 2.

Hence,  $A'$  has coordinates  $(8, -2)$ .

11. The graph of  $y = x \ln x^2$  is shown.  
How many solutions does the equation  $|x \ln x^2| = \frac{1}{3}(x+1)$  have?

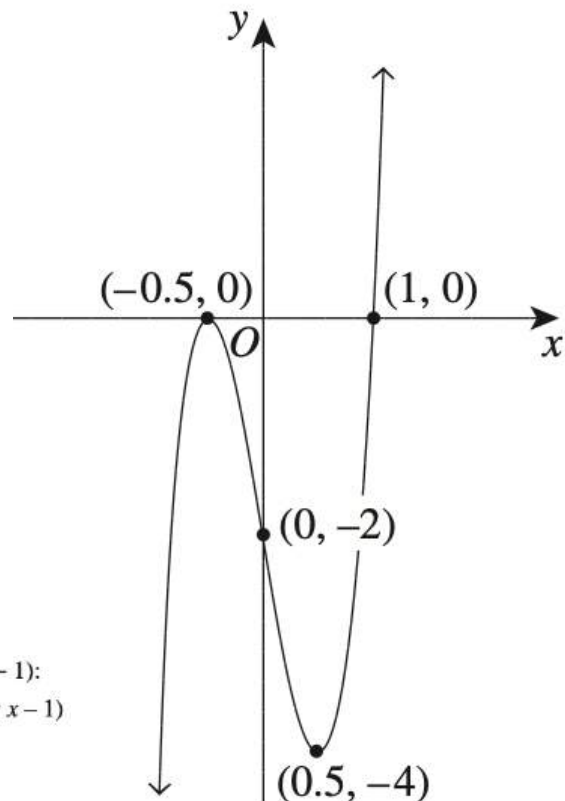
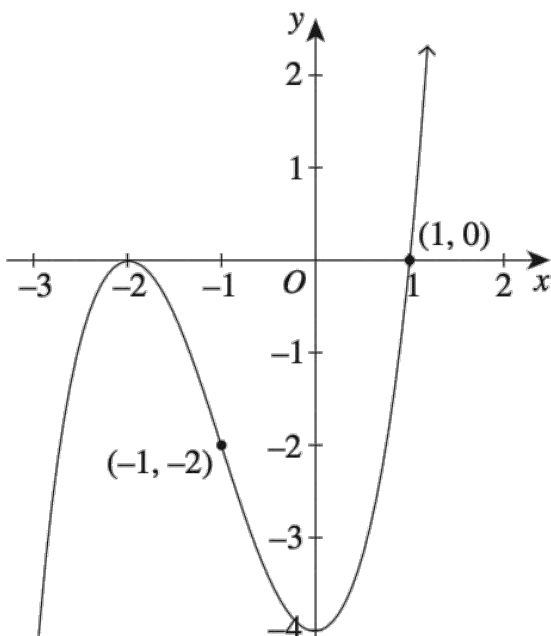
Therefore, there are five points of intersection.



12. The diagram shows the graph of  $y = f(x)$ .

3

Sketch the graph of  $y = f(2x - 1)$ , labelling all stationary points and intercepts.



$f(x)$  undergoes the following transformations to become  $y = f(2x - 1)$ :

- horizontal translation of 1 unit to the right ( $x$  is replaced by  $x - 1$ )
- horizontal dilation by a factor of  $\frac{1}{2}$  ( $x$  is replaced by  $2x$ )

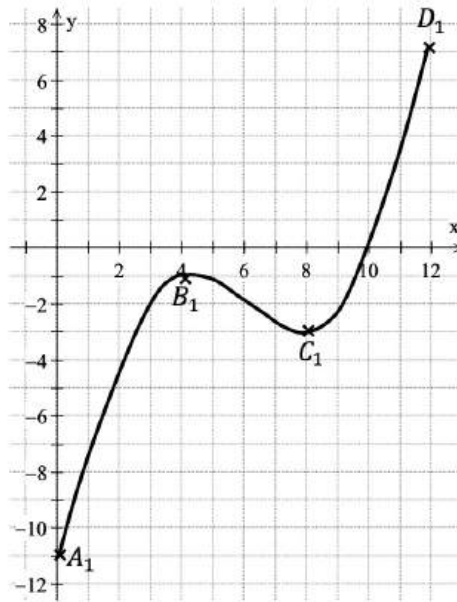
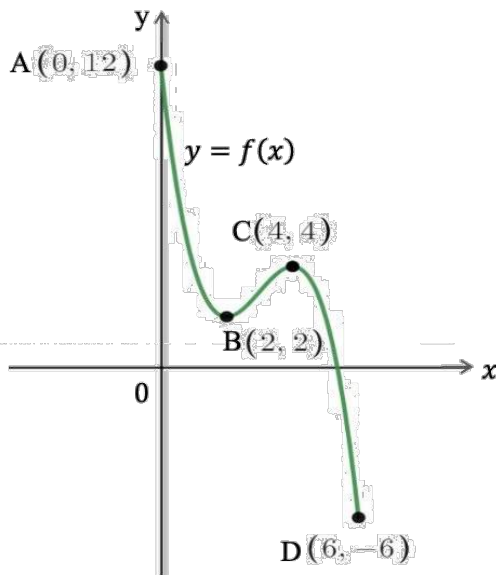
Therefore, sketching the graph of  $y = f(2x - 1)$  gives:



13. The diagram shows the graph of the function  $y = f(x)$ .

Sketch the graph of the function  $y = 1 - f\left(\frac{x}{2}\right)$  showing the coordinates of the turning points and the  $y$  intercepts on the number plane provided below.

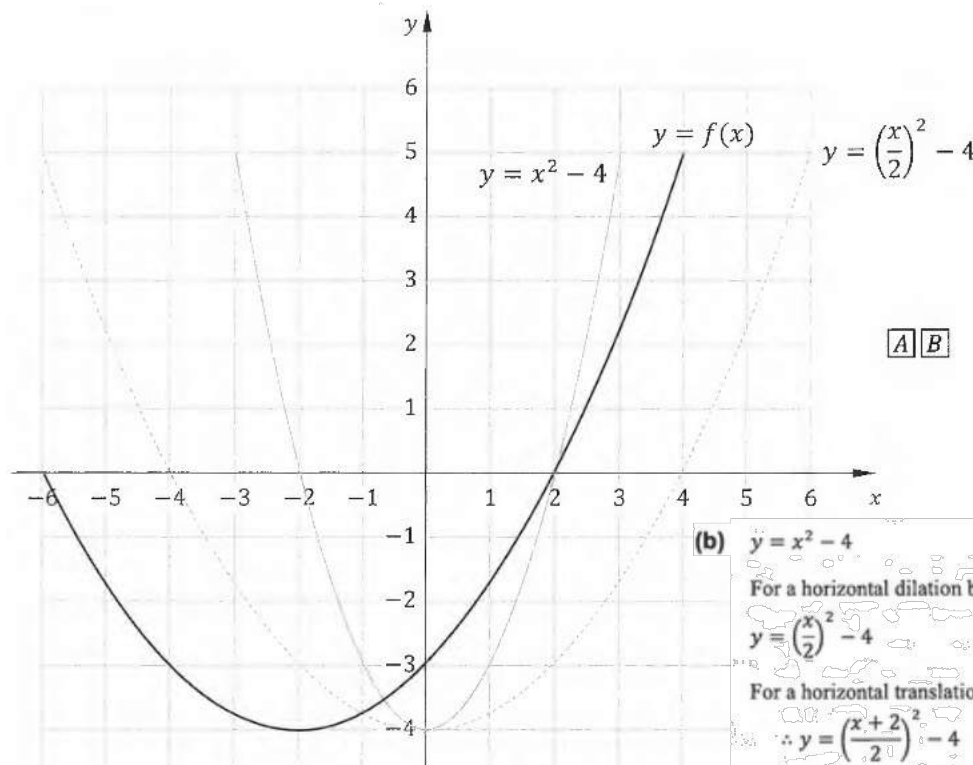
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14. Consider the graph of  $y = x^2 - 4$  as shown.

The function  $f(x)$  is the function  $x^2 - 4$  horizontally dilated by a factor of 2, then translated left 2 units.

- (a) Sketch the graph of  $y = f(x)$  on the axes above, clearly showing all intercepts. 2  
 (b) Write the function  $f(x)$  in the form  $f(x) = (a(x + b))^2 - 4$ . 2



(b)  $y = x^2 - 4$

For a horizontal dilation by a factor of 2 replace  $x$  with  $\frac{x}{2}$

$$y = \left(\frac{x}{2}\right)^2 - 4$$

For a horizontal translation 2 units to the left replace  $x$  with  $x + 2$

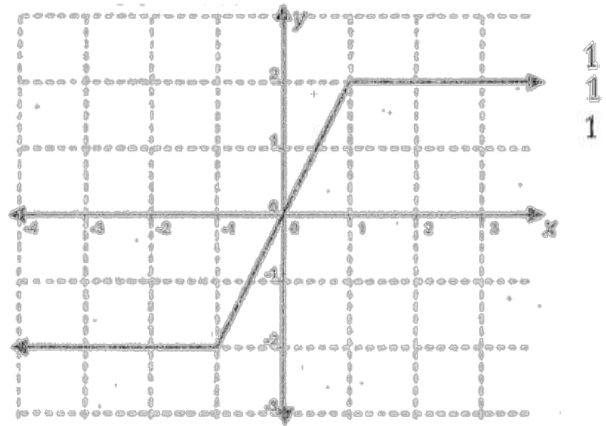
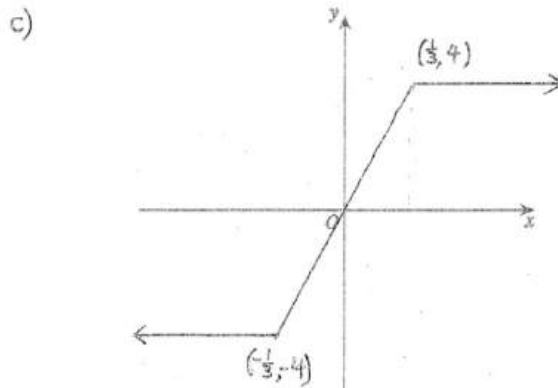
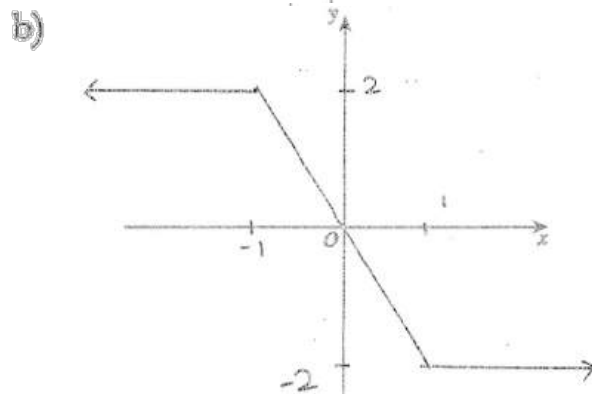
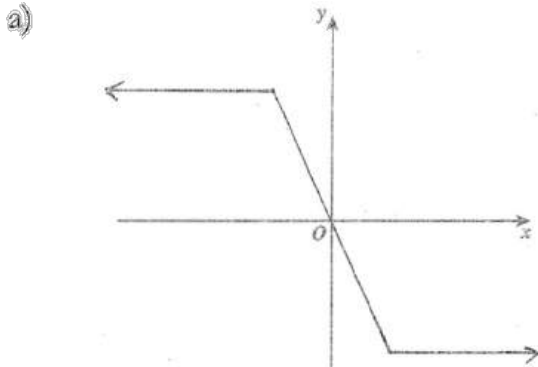
$$\therefore y = \left(\frac{x+2}{2}\right)^2 - 4$$

$$\therefore f(x) = \left(\frac{1}{2}(x+2)\right)^2 - 4 \quad \therefore a = \frac{1}{2}, b = 2$$

15. The graph of a function  $y = \frac{4}{x+1}$  is translated 3 units left, and dilated vertically by a factor of  $\frac{1}{2}$ . Which of the following gives the equation of the new function?
- A)  $\frac{y}{2} = \frac{4}{x-2}$     B)  $2y = \frac{4}{x-2}$     **C)  $2y = \frac{4}{x+4}$**     D)  $\frac{y}{2} = \frac{4}{x+4}$

16. Consider the graph  $y = f(x)$  shown below.

- a) Sketch the graph of  $y = -f(x)$   
 b) Sketch the graph of  $y = f(-x)$   
 c) Sketch the graph of  $y = 2f(3x)$



17. The function  $f(x) = x^2$  is transformed into a new function whose graph is shown below:  
 Find the equation of the new function in the form  $g(x) = k(x+b)^2 + c$  for some constants  $k$ ,  $b$ , and  $c$ .

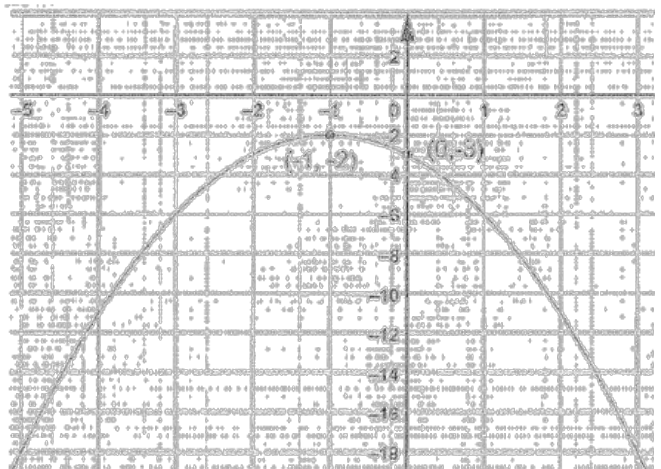
$$g(x) = k(x+1)^2 - 2$$

$$g(0) = -3$$

$$k(1)^2 - 2 = -3$$

$$k = -1$$

So  $g(x) = -(x+1)^2 - 2$



1. A function  $y=f(x)$  undergoes the following transformations:

3

- A reflection in the  $x$  axis
- Then a horizontal dilation with scale factor  $\frac{1}{3}$
- Then a horizontal translation 7 units left

The point  $P(x, y)$  lies on  $y=f(x)$ . The image of  $P$  on the transformed function is  $P'(-2, 1)$ . Find the coordinates of  $P$ .

$$\begin{aligned} a=3, b=7, k=-1 \checkmark \\ x_2 &= \frac{1}{a}x - b \\ -2 &= \frac{1}{3}x - 7 \\ 5 &= \frac{1}{3}x \\ x &= 15 \\ \therefore P(15, -1) \checkmark \end{aligned}$$

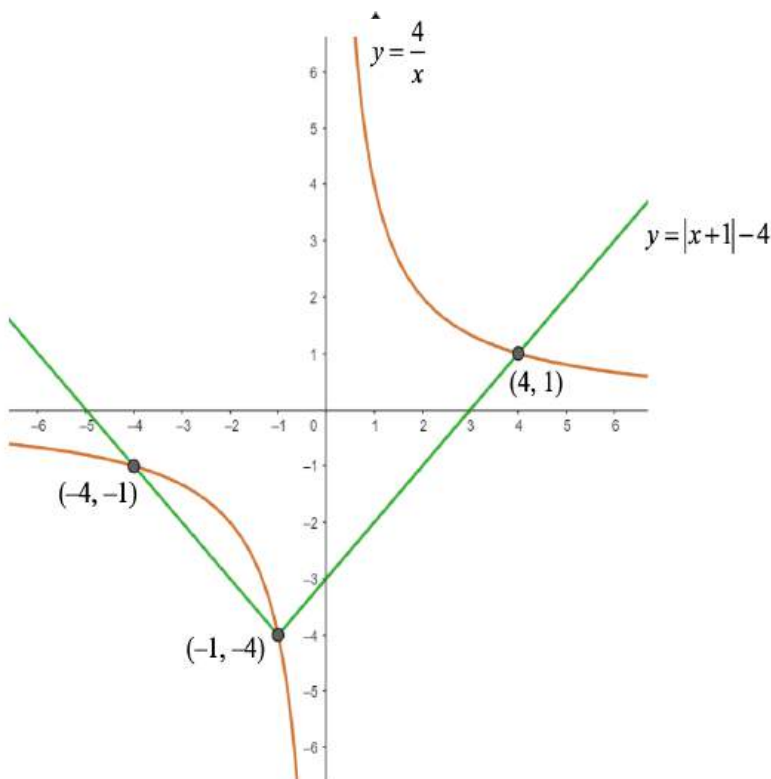
$$\begin{aligned} y_2 &= ky + c \\ 1 &= -y \\ y &= -1 \end{aligned}$$

2. a) Sketch the graphs of the functions  $y = \frac{4}{x}$  and  $y = |x+1| - 4$  on the number plane below. Clearly show their point(s) of intersection and any intercepts on the axes.

4

- b) Hence, or otherwise, solve  $|x+1| - 4 \leq \frac{4}{x}$

2



b)  $[-4, -1] \cup (0, 4]$

3. (a) Find the vertical and horizontal asymptotes of  $y = \frac{3x-3}{(x+3)(x-2)}$ . 2
- (b) Hence sketch the graph of  $y = \frac{3x-3}{(x+3)(x-2)}$ , showing all intercepts and asymptotes. 3
- (c) Hence or otherwise, find all values  $x$  for which  $\frac{3x-3}{(x+3)(x-2)} \geq 1$ . 3

a.) Vertical at  $(x+3)(x-2) = 0$

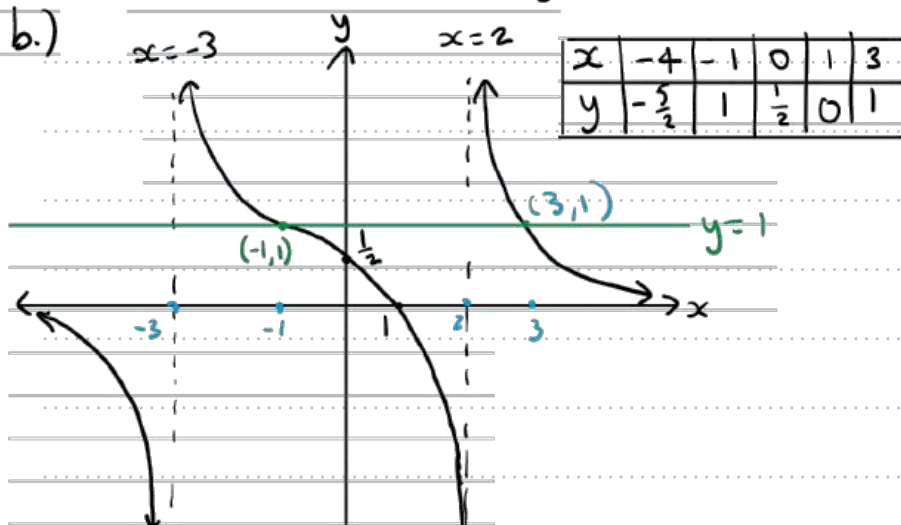
$\therefore x = -3, 2$

$$y = \frac{(3x-3)}{(x^2+x-6)} \div x^2$$

$$= \frac{\frac{3}{x} - \frac{3}{x^2}}{1 + \frac{1}{x} - \frac{6}{x^2}}$$

$\therefore y = 0$  as  $x \rightarrow \pm\infty$

Asymptotes at  $x = -3, 2$  and  $y = 0$



c.) Let  $\frac{3x-3}{(x+3)(x-2)} = 1$

$$3x-3 = (x+3)(x-2)$$

$$= x^2 + x - 6$$

$$0 = x^2 - 2x - 3$$

$$= (x-3)(x+1)$$

$\therefore x = 3, -1$  intersect (see green line)

$-3 < x \leq -1$  and  $2 < x \leq 3$

$(-3, -1] \cup (2, 3]$

4. Which transformations listed are required to obtain the graph of  $y = x^2 + \frac{1}{2}x + 3$  from the graph of  $y = 4x^2 + x$ ?

A. Horizontal dilation by a factor of 2; vertical translation of 3 units upwards

B. Horizontal dilation by a factor of  $\frac{1}{2}$ ; vertical translation of 3 units to downwards

C. Vertical dilation by a factor of 2; horizontal translation of 3 units in to the right

D. Vertical dilation by a factor of  $\frac{1}{2}$ ; horizontal translation of 3 units to the left

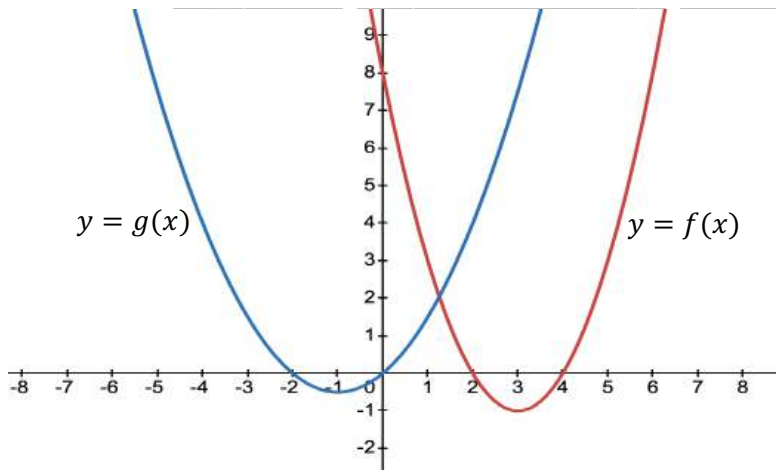
If  $f(x) = 4x^2 + x$ , then  $f\left(\frac{1}{2}x\right) + 3 = 4\left(\frac{1}{2}x\right)^2 + \left(\frac{1}{2}x\right) + 3 = x^2 + \frac{1}{2}x + 3$

The graph of  $y = f\left(\frac{1}{2}x\right) + 3$  is obtained from the graph of  $y = f(x)$  through a horizontal dilation of factor 2, and a vertical translation of 3 units upwards.



5. Let  $f(x) = (x-3)^2 - 1$ .  $f(x)$  is transformed to  $g(x)$  such that  $g(x) = \frac{1}{2}f(2-x)$ .

- (a) Complete a possible sequence of transformations that maps  $f(x)$  to  $g(x)$ . 2  
 (b) On the same set of axes below, sketch the graph of  $y = g(x)$  below. 2



- a) Reflect about the y-axis  
 translate 2 units to the right  
 vertical dilate by factor of  $\frac{1}{2}$

6. Let  $f(x) = x^2 - 4x$ .

Sketch the graph of  $y = 2f(1-x) + 6$ , showing the location of the vertex and the intercepts. 3

Since  $2f(1-x) + 6 = 2f(-(x-1)) + 6$ , our graph will  
 be  $f(x)$  transformed as follows: - reflect in y-axis;  
 Shift Right by 1; dilate by factor 2 vertically; shift up by 6.  
 Hence new vertex is  $(3, -4) \rightarrow (-1, -2)$

Easiest is just to transform algebraically:-

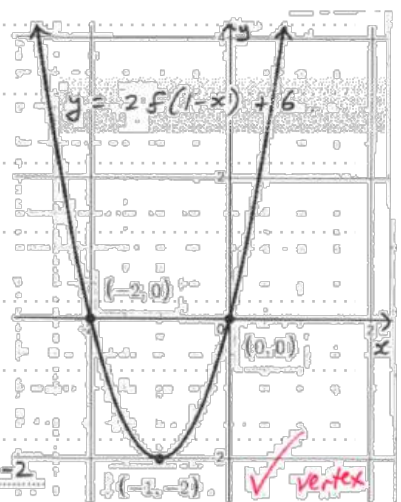
$$y = 2f(1-x) + 6 = 2[(1-x)^2 - 4(1-x)] + 6$$

$$\therefore y = 2[1 - 2x + x^2 - 4 + 4x] + 6 = 2(x^2 + 2x - 3) + 6$$

$$\therefore y = x^2 + 2x = x(x+2) \quad \rightarrow \text{x-ints are } 0, -2$$

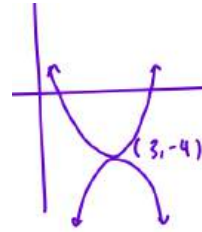
$$\& \text{ axis of } x = -1$$

$$\therefore V = (-1, -2)$$



7. Given  $f(x)$  is a quadratic function with a vertex at  $(3, -4)$ , which of the following must be true?

- A.  $f(x) = 0$  has one or more integer roots.  
B.  $f(x) - 3 = 0$  has one or more rational roots.  
**C.  $f(x) + 4 = 0$  has one or more real roots.**  
D.  $f(x) + 5 = 0$  has no real roots.



8. Consider the function  $f\left(\frac{x}{3}-1\right) = \frac{1}{x-2} - 6$ .

- (a) List a set of transformations that, when applied in order, would transform  $f(x)$  to  $f\left(\frac{x}{3}-1\right)$ . 2  
(b) Find the equation of  $y = f(x)$ . 2

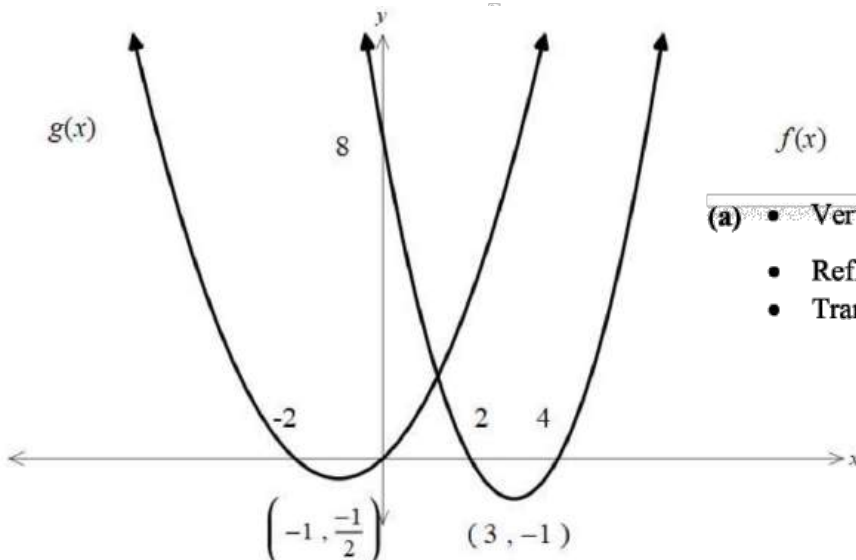
(a)  $f\left(\frac{1}{3}(x-3)\right)$  ① dilate horizontally by a factor of 3  
② translate right by 3 units

(b)  $f(x-1) = f\left(\frac{3x}{3}-1\right)$  dilate horizontally by a factor of  $\frac{1}{3}$   
 $= \frac{1}{3x-2} - 6$  ✓

$f(x) = f((x+1)-1)$  translate left by 1 unit  
 $= \frac{1}{3(x+1)-2} - 6 = \frac{1}{3x+1} - 6$

9. Let  $f(x) = (x-3)^2 - 1$ . A transformation maps  $f(x)$  to  $g(x)$  such that  $g(x) = \frac{1}{2}f(2-x)$ .

- (a) Complete a possible sequence of transformations that maps  $f(x)$  to  $g(x)$ . 2  
(b) On the same set of axes sketch the graph of  $g(x)$  below. 2

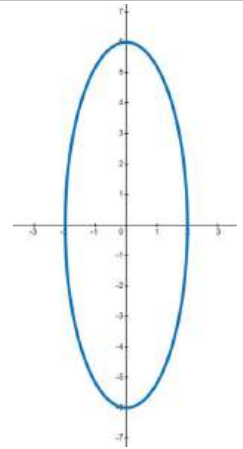


- (a) • Vertical dilation of  $\frac{1}{2}$   
• Reflection in  $y$ -axis  
• Translation of 2 units in the positive  $x$ -axis

10. Which of the following best represents the graph of  $(3x)^2 + y^2 = 36$ ?

Horizontal dilation by a factor of  $\frac{1}{3}$

Circle is compressed horizontally making the left/right extremes at  $\pm 2$  now instead of  $\pm 6$ .



11. The graph of the function  $f(x) = x^3$  is translated up 9 units, left 4 units, then dilated vertically by a scale factor of  $\frac{1}{3}$ , to make a new graph,  $g(x)$ .

- (a) Find the equation,  $g(x)$ , of the new graph. 3  
 (b) A point  $P(a, b)$  on  $f(x)$  is transformed to the point  $Q(-1, 12)$  on  $g(x)$ . 2  
 Find the values of  $a$  and  $b$ .

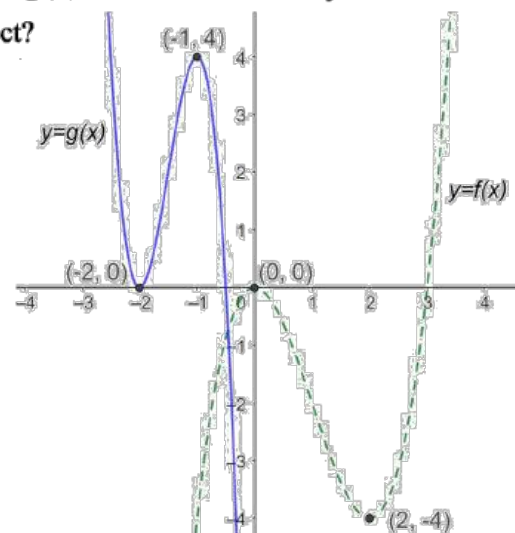
(a)  $f(x) = x^3$   
 $f(x) + 9$   
 $f(x+4) + 9$   
 $\frac{1}{3}[f(x+4) + 9]$

$g(x) = \frac{1}{3}((x+4)^3 + 9)$   
 $= \frac{1}{3}(x+4)^3 + 3$

(b) Use inverse operations  
 $(-1, 12)$   
 $\downarrow \times 3$  (inverse of dilate vertically  $\frac{1}{3}$ )  
 $(-1, 36)$   
 $+4 \downarrow$  (inverse of translate left 4 units)  
 $(3, 36)$   
 $\downarrow -9$  (inverse of translate up 9 units)  
 $(3, 27)$

12. The graphs of  $y = f(x)$  and  $y = g(x)$  are shown. The function  $g(x)$  can be obtained by applying a transformation to  $f(x)$ . Which of the following is correct?

- A.  $g(x) = -f(x+4)$   
 B.  $g(x) = -f(2x+4)$   
 C.  $g(x) = f(-2x)+4$   
 D.  $g(x) = f(-x+2)+4$



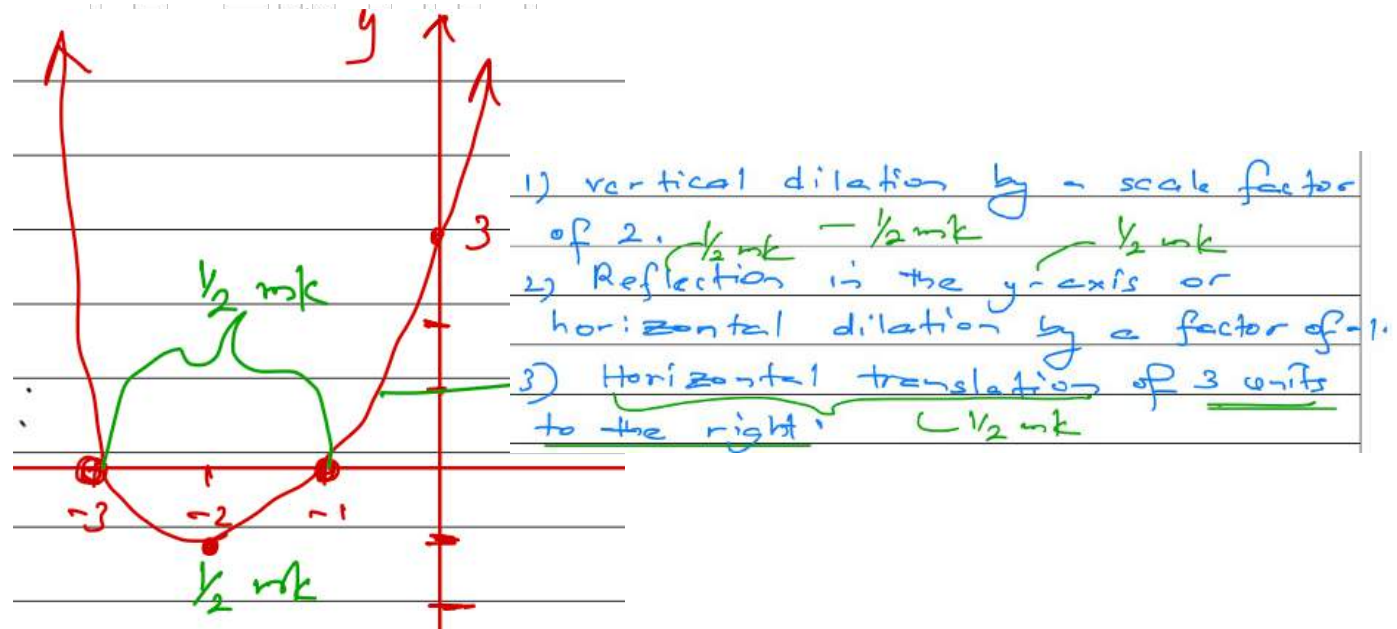
Graph of  $g$  can be obtained from  $f$  by applying:

A horizontal dilation by a factor of  $\frac{1}{2}$ :  $f(2x)$

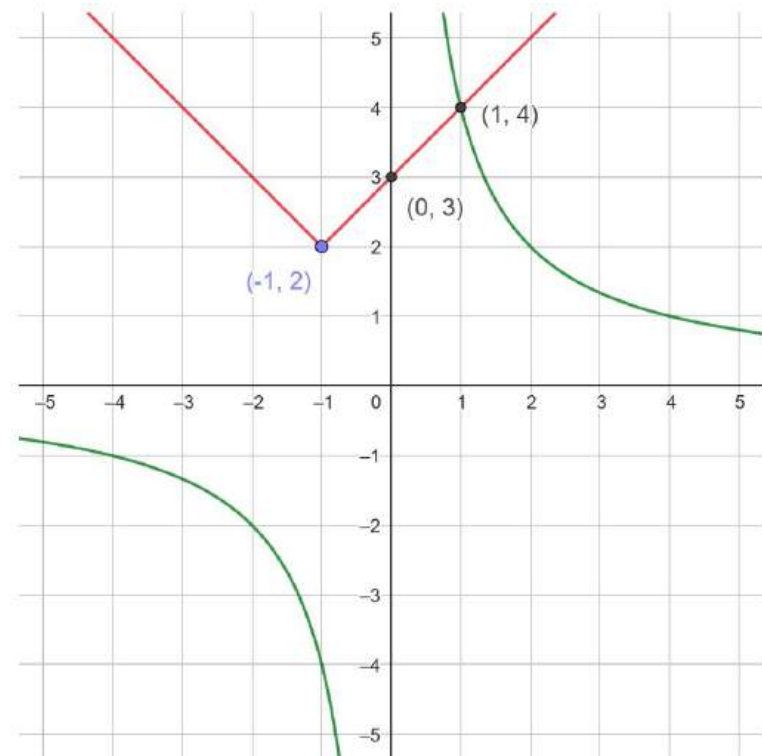
Translation by 2 units to left:  $f(2(x+2))$

Reflection on x-axis:  $-f(2x+4)$

13. (a) Let  $f(x) = (x+2)^2 - 1$  2  
Plot the graph of  $f(x)$  in the given space below, showing all intercepts and turning point.
- (b)  $f(x)$  is transformed to  $g(x)$  such that,  $g(x) = 2f(3-x)$  3  
Describe the three transformations in the correct order.



14. (a) Sketch the graph of the functions  $y = \frac{4}{x}$  and  $y = |x+1| + 2$ , showing coordinates of their intersection point and intercept(s). 4
- (b) Hence, or otherwise, solve  $\frac{4}{x} \geq |x+1| + 2$ . 1



$$0 < x \leq 1$$